

Task 1 – Order and Sales Analysis

a) The first query extracted the total count of orders grouped by status: **Pending**, **Delivered**, and **Shipped**. This gives a high-level snapshot of the current distribution of orders across various stages in the fulfillment pipeline.

b) Next, I calculated the monthly ratio of fulfilled orders (Shipped + Delivered) to Pending orders.

This is a valuable operational KPI:

- An increasing ratio suggests improving efficiency in moving orders from pending to fulfillment.
- A declining or stagnant ratio highlights delays or blockages in the order processing pipeline.
By tracking this monthly, the company can measure whether order fulfillment is keeping pace with new order inflow.

c) To get a clearer view of revenue trends, I wrote queries that calculate monthly sales separately for each order status (**Delivered**, **Shipped**, and **Pending**).

- Delivered sales reflect actualized revenue.
- Shipped indicates in-transit value.
- Pending captures potential revenue stuck in the pipeline.

d) I then calculated the percentage distribution of order statuses for each month.

This reveals whether a significant portion of monthly orders remain in **Pending** status — useful for diagnosing processing inefficiencies, especially when tracked over time.

(data cleansing) NOTE - While analyzing the joined dataset (**orders_payments**), I implemented a data validation query to check the logical sequence of order and payment dates. Specifically, I filtered for records where the **payment date occurred before the order date**, which should not happen in any standard transaction flow.

This query returned 7454 rows that revealed a **critical data inconsistency** — payments being recorded **before** the corresponding orders were placed.

Task 2 – Customer Analysis

a) The first query identified repeat customers, i.e., those with more than one order. These customers are likely more satisfied and engaged.

This segment can be prioritized with reward programs, early access, or premium support to encourage continued loyalty.

b) I analyzed monthly unique customer counts to determine which months were most successful in acquiring new users.

This insight helps the marketing and outreach teams identify what worked during high-growth periods — such as campaigns or partnerships — and replicate those strategies.

c) I then segmented customers based on total payment amounts per order:

- **Low Spenders** (< ₹1,000)
- **Medium Spenders** (₹1,000–1,200)
- **High Spenders** (> ₹1,200)

This segmentation helps the finance and sales teams personalize offers, recommend suitable lease plans, or prioritize support for high-value users.

d) A critical analysis was calculating the customer retention period — the number of days between each customer's first and most recent order.

This metric:

- Helps the company identify loyal, long-term customers who can be nurtured with loyalty rewards.
 - Flags low-retention users, allowing the outreach or CRM team to re-engage them with targeted campaigns or offers.
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Task 3 – Payment Status Analysis

a) The first query provided a summary of total payments grouped by status (**Completed**, **Pending**, **Failed**), including the net total collected per category. This serves as a top-level financial overview to assess payment health and identify where revenue may be getting blocked.

b) Next, I analyzed monthly payment trends by status.

This shows whether pending or failed payments are increasing in specific months — possibly indicating technical issues with the payment gateway, poor follow-ups, or system bugs.

c) I identified high-risk customer accounts — those with a high percentage of failed payments.

This list can be sent to the finance team for flagging or double-checking during approval, as these users may pose a financial risk or signal fraudulent behavior.

d) Finally, I calculated the overall ratio of completed payments to (pending + failed) payments, which came out to ~0.50.

This is concerning, as it means only 1 out of every 2 payment attempts is successful. The rest are still pending or failed, implying:

- Assets have been deployed but revenue hasn't been realized.
- There may be inefficiencies or gaps in payment collection, system design, or follow-up procedures.

This finding strongly suggests a need for the business to optimize payment processes, reduce friction, and possibly introduce automated reminders or follow-up mechanisms.

Task 4 – Order Details Report

a) I began this task by creating a **view named `orders_payments`**, which joins the **`orders`** and **`payments`** tables on the common column **`order_id`**.

This consolidated table allowed me to perform deeper, cross-functional analysis combining order, payment, customer, and shipping data.

b) The first query from the view calculated the **total number of distinct orders**, the **total revenue**, and the **payment success rate** (i.e., percentage of completed payments).

This gives a complete snapshot of overall transaction performance and how effectively the business is converting orders into revenue.

c) Then, I tracked the **cumulative revenue per customer** on a monthly basis.

This helps the company understand **long-term customer value trends** and identify high-growth customer accounts based on their order history.

d) I filtered and listed **high-value customers** — those whose total spending exceeds ₹2000.

This is helpful for the **sales and retention teams**, who can offer these customers loyalty rewards, exclusive offers, or premium support to boost satisfaction and retention.

e) I calculated **monthly total orders and total sales**, giving a clear overview of business activity and revenue trends across months.

This helps in identifying **peak and low sales periods**, assisting with inventory, staffing, and budgeting decisions.

f) I then measured the **Average Order Value (AOV)** per month.

Tracking this metric helps the company assess **whether upselling strategies or higher-tier product offerings** are effective over time.

g) I estimated the **potential lost revenue** from orders with **pending** or **failed** payment statuses.

This reveals how much money is being held up due to unsuccessful transactions, signaling the need for payment system improvements or customer follow-up mechanisms.

h) I calculated the **average payment delay** for each customer by measuring the time gap between order date and actual payment date.

Customers with high delays might need automated reminders or pre-billing arrangements to improve cash flow consistency.

i) I identified the **top 10 most frequently used shipping addresses**.

This helps Alt Mobility understand regional demand patterns, optimize delivery logistics, and plan for local marketing or expansion strategies.

j) Lastly, I detected records where **payment was made before the order was placed**, a logical error in data.

Flagging such cases supports **data cleaning efforts** and improves the overall quality and reliability of analytics going forward.

FINDINGS FROM CUSTOMER RETENTION REPORT

Conclusion and Insights - for Bar Graph

- Initial Retained Customer Trend: The bar chart clearly shows a decreasing trend in initial retained customers over time. Cohorts from early 2020 (e.g., 2020-01) had significantly higher initial retained customer counts compared to more recent cohorts (e.g., 2025-04).
- Peak Initial Retention: The cohort of 2020-01 had the highest initial retained customer count, with 241 customers.
- Lowest Initial Retention: The cohort of 2025-02 and 2025-04 had the lowest initial retained customer count, with 39 customers.

Conclusion and Insights - for Line Chart

- **Initial Customer Loss:** There is a substantial initial drop in retained customers, as indicated by the decay rate of -97.18% in the first month .
- **Fluctuating Retention:** The decay rate varies significantly over time, suggesting that customer retention is influenced by multiple factors that change from month to month .
- **Long-Term Growth:** Despite the fluctuations, the average number of retained customers shows a general increasing trend over the observed period .

Recommendations for Alt Mobility on basis of queries results

1. Improve Fulfillment Efficiency

- **Insight:** The ratio of fulfilled (shipped/delivered) to pending orders fluctuates and in some months is low.
 - **Recommendation:**
 - Investigate and resolve delays in the order-to-fulfillment pipeline.
 - Automate alerts for orders stuck in **Pending** for too long.
 - Analyze logistics partner performance if applicable.
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2. Recover Potential Lost Revenue

- **Insight:** A significant portion of orders have **pending** or **failed** payments, with a **success ratio of only ~50%**.
- **Recommendation:**
 - Implement automated payment reminders and retry mechanisms.

- Introduce wallet-based payments or instant UPI options for better conversion.
 - Consider pre-authorization or tokenized payment setups to reduce failures.
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3. Prioritize High-Value and Loyal Customers

- **Insight:** Certain customers consistently place large orders or have high lifetime value.
 - **Recommendation:**
 - Create loyalty programs or perks for high-value customers.
 - Assign account managers to VIP customers for long-term relationship building.
 - Use retention cohorts to proactively engage users likely to churn.
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4. Leverage Successful Months for Customer Acquisition

- **Insight:** Some months saw a spike in new, unique customers.
- **Recommendation:**
 - Identify what campaigns or strategies were in place during high-growth periods.
 - Replicate or scale those initiatives (ads, referral bonuses, local partnerships).

- Use that month as a benchmark for goal setting.

Recommendations for Alt Mobility on basis of visualisation result

Retained customers per cohort have been **steadily declining** since 2020.

This drop indicates **weaker customer stickiness** or satisfaction in newer users.

Launch Loyalty and Reward Programs

- Retention is stronger in older cohorts—likely due to longer-term value buildup.
- Introduce referral bonuses, milestone rewards, or points-based systems for newer users.

Improve Post-Purchase Engagement

Send proactive reminders, usage tips, and follow-up messages after order completion to increase repeat activity.

Collect Feedback from Recent Churned Users

Run surveys or interviews to understand why new users are not returning.

